

AGORA

One Sentence, One Living World

Generating Persistent Worlds for Human–AI Social Interaction

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Abstract. Generating a social world requires more than a coherent scene: inhabitants must be able to change shared spatial, material, and social state. We present Agora, a system that compiles a one-sentence premise into a persistent rendered environment with rooms, roles, institutions, possessions, private knowledge, and visual identity. Humans and language agents use a generated action repertoire or propose context-specific actions; a World Coordinator evaluates their effects before updating shared state. Across ten paired world briefs, compositional generation completes 10/10 worlds versus 7/10 for a single-pass baseline, at higher token and latency cost. Five two-round interaction runs produce 53 events, ten approved open proposals, and thirteen events targeting the human interactor; 49/56 adjudicated action results succeed. A separate retrospective audit covers 24 one-sentence generation trials and 21 scripted 24-round trajectories: 1,365/1,512 model actions execute successfully, yet only 398 successes change inventory or location. The longer harness uses centralized planning, and its relation updates mostly repeat a fixed rule. These observations distinguish construction and execution reliability from behavioral fidelity or causal social effects. We release reproducible figures, bilingual artifact-evaluation materials, and a prospective memory-intervention protocol. Agora connects language-based world authoring to inspectable trajectories of what inhabitants do together.



Figure 1. Five generated settings share a world schema while varying institutions, spaces, and visual identity. Panels show preserved map assets and representative character frames; they are not new experimental outputs.

1 Introduction

A generated social world needs persistent consequences. In a floating embassy where endangered languages are political assets, inhabitants must be able to access archives, trade translation rights, and respond when a flood threatens the only surviving treaty. These activities require explicit places, possessions, information, and rules behind the rendered scene.

Several research lines address parts of this problem. Procedural content generation studies representations and criteria for functional game content [1, 2]. Word2World and DreamGarden extend language-based authoring to playable games and development plans [3, 4]. Generative Agents demonstrates how local memory and encounters can produce community activities in an authored town [5], while Concordia grounds language-described actions through a Game Master [6]. These advances motivate a joint design question: how can one short premise specify both a world and the conditions under which a society can act within it?

We present Agora, a system that compiles a one-sentence premise into a persistent, visually grounded environment for humans and language agents. Its representation connects rooms, roles, institutions, possessions, private knowledge, visual assets, and action opportunities. A generated repertoire

makes the world’s activities discoverable; an open-proposal channel lets an actor express a more specific intention. A World Coordinator interprets each proposal against shared constraints before its effects enter the world state. For example, proposing a treaty exchange does not by itself transfer ownership: the resources, participants, and required acceptance must support the change.

The research object is a state trajectory

$$\tau = (S_0, q_1, d_1, S_1, \dots, q_T, d_T, S_T), \quad (1)$$

where q_t is an intended action and d_t is the coordinator decision. This representation distinguishes an intention, an enacted effect, and a later response. It also makes explicit which observations concern software execution, simulated social behavior, or human experience.

Our contributions are threefold:

1. A compositional language-to-world pipeline connecting semantic, visual, spatial, and mutable social state, with dependency-directed local refinement.
2. A shared action interface for humans and agents that combines generated opportunities with coordinator-governed proposals and persistent consequences.
3. Audited evidence from ten paired world briefs, a separate

24-cell one-sentence generation corpus, five locally observed interaction runs, and 21 scripted execution trajectories, distinguishing successful calls from material consequences and retained failure cases.

The completed studies characterize an authoring and experimentation platform. A paired memory-intervention design remains prospective (Appendix E); behavioral fidelity and causal social effects require further evidence.

2 Related Work and Research Position

2.1 From Generated Content to Executable Worlds

Procedural content generation distinguishes what is represented, how it is generated, and how its fitness is evaluated [1]. PCGML further connects learned content models with repair and critique [2]. This lineage matters for Agora: decomposing a world and repairing a faulty part are established design ideas. Our use of them ties a repair to explicit dependencies among role, inventory, geometry, media, and runtime state.

Word2World converts stories into playable tile-based games and evaluates its generation stages [3]. DreamGarden recursively decomposes a user prompt into plans executed by specialized modules, with user feedback during Unreal Engine development [4]. Both are close precedents for prompt-driven, staged authoring. Agora emphasizes a reusable social-state representation: the generated institutions and objects become inputs to ownership, knowledge access, and action adjudication during play. We make no claim that prompt-based world generation or hierarchical planning is new.

Environment generation also serves embodied-agent training. ProcTHOR samples diverse interactive houses [7]; Holodeck uses language and spatial constraints to produce 3D scenes from prompts [8]. Genie instead learns action-controllable visual dynamics from video [9]. These approaches provide distinct substrates for interaction. Agora uses explicit entity identities and state transitions in a rendered 2D environment, making social and material changes directly inspectable. Visual generation and symbolic persistence address complementary aspects of world construction.

2.2 Grounded Dialogue and Shared Play

TextWorld provides generated text games with explicit state tracking and controllable task difficulty [10]. ScienceWorld evaluates whether agents can carry out scientific procedures in an interactive environment [11]. Both motivate evaluating executed consequences rather than the plausibility of a textual plan.

LIGHT is a particularly relevant predecessor: humans and models inhabit a fantasy environment, combining dialogue with actions grounded in local objects and characters [12]. Thus shared human-agent play and situated dialogue are not unique to Agora. Our focus is generating the surrounding social and material setting from a premise, then coupling visible activities and novel proposals to the same persistent state. Human-directed events in our logs establish reachability of the player, while experience and collaboration quality require a participant study.

2.3 Persistent Communities and Social Simulation

Generative Agents combines observations, retrieval, reflection, and planning in a town of 25 agents, illustrating how remembered encounters can support coordinated activities [5]. Concordia provides a flexible simulation framework in which agents describe intentions and a Game Master grounds their effects [6]. The World Coordinator follows this separation of intention and environment response; Agora connects it to premise-generated identities, inventories, rules, and rendered spaces.

AgentSociety studies large populations and interventions in a societal environment [14]; LMAgent combines multi-modal shopping behavior with large-scale social interaction [15]. Avalon studies cooperation and confrontation under explicit game rules [16]. These systems show why trajectories and institutions matter beyond isolated dialogue quality. Our smaller studies address authoring diversity and trace inspectability, without claiming comparable scale or empirical correspondence to society.

2.4 Action, Feedback, and Feasibility

ReAct interleaves reasoning and environment actions [17]; Reflexion stores verbal feedback for subsequent attempts [18]; Voyager accumulates executable skills through environment feedback and verification [19]. These methods motivate the observation–intention–outcome cycle. SayCan explicitly connects language-level usefulness to skill feasibility [20]. Agora applies the analogous distinction to social and material effects: an intention can be intelligible while its requested transfer, construction, or status change is infeasible. Its action interface adds partner acceptance, ownership, and world-specific rules to that decision. This is a system design connection, not a claim of algorithmic equivalence or an empirical comparison with those agents.

2.5 Evaluation and Validity

SOTOPIA evaluates interaction through multiple social goals and criteria [13]. Subsequent work shows that an omniscient simulation can appear more successful than an information-asymmetric interaction [21]. This motivates our explicit observation boundary and private knowledge, but does not by itself validate their implementation. SOTOPIA- π also reports evaluator overestimation of agents trained for social interaction [22]. Together with broader LLM-judge biases [23], this motivates keeping structural checks, state-derived metrics, and future human ratings separate.

AgentBoard motivates inspecting intermediate progress rather than relying on final success alone [24]. We similarly report executed changes separately from plausible intentions, but do not claim that our mutation counts measure validated task progress. AI Agents That Matter emphasizes cost, hold-outs, and reproducibility [25]; this informs our separation of unequal-budget architecture comparisons from model coverage audits and our explicit treatment of retrospective evidence.

The ODD protocol emphasizes explicit entities, scheduling, initialization, and model rationale for reproducible agent-based modeling [26]. We adopt these reporting priorities in Appendix A. Human-grounded agent simulation, such as mod-

Research line	Representative substrate	Action or generation mechanism	Relation to Agora
Word2World; DreamGarden [3, 4]	Generated game content and environments	Story expansion; hierarchical development plans	Precedents for language-driven, staged authoring
ProcTHOR; Holodeck [7, 8]	Procedural or language-conditioned 3D scenes	Interactive objects and spatial constraints	Physical environment diversity and grounding
Genie [9]	Learned visual dynamics	Latent, frame-level control	Complementary representation of world dynamics
LIGHT; SOTOPIA [12, 13]	Grounded fantasy play; social scenarios	Dialogue, actions, and social goals	Shared interaction and social evaluation
Generative Agents; Concordia [5, 6]	Persistent communities; grounded simulations	Memory and planning; Game Master adjudication	Direct precedents for memory and action grounding
AgentSociety; LMAgent [14, 15]	Societal and e-commerce environments	Large populations and domain-specific interactions	Scale and behavioral evaluation beyond this study
Agora	Premise-generated 2D social worlds	Typed shared state; repertoire and open proposals	Connects world authoring to inspectable social trajectories

Table 1. Research positioning by representation and mechanism. Entries describe the cited systems’ emphasis, not an exhaustive feature audit or a performance ranking. We do not treat unreported capabilities as absent.

els built from interviews and surveys and evaluated against the people they represent [27], sets a different evidentiary standard from fictional world coherence. Finally, work on complex contagion distinguishes information transmission from socially reinforced adoption [28]. Accordingly, the prospective intervention measures both warning diffusion and material treaty rescue; neither network density nor fluent conversation substitutes for coordinated action.

3 The Agora System

3.1 World Representation

Given a one-sentence premise p , the generation pipeline constructs

$$W = (E, G, R, K, A, M, P), \quad (2)$$

where E contains entities and their stable identities; G is traversable geometry; R contains physical, economic, and institutional rules; K specifies knowledge allocation and access; A is the initial action repertoire; M associates visual media with entities; and P records world identity and revision. Mutable state S_t contains locations, holdings, knowledge acquired during play, tasks, relationships, and event status. The schema is shared across worlds; its entities and rules depend on p .

This separation gives language-generated descriptions operational meaning. An archive must refer to an existing room, an item must have a valid owner, and a private fact must have a defined holder. Each visual asset is linked to the same entity identity used by the runtime. Changes then propagate through the state observed by subsequent actors rather than remaining only in dialogue.

3.2 Compositional Generation and Local Refinement

A planning stage identifies institutions, conflicts, resources, and activity loops. Specialized stages expand materials, rooms, items, roles, main characters, knowledge, visual direction, wardrobe, property, and placement. The shared representation constrains references between their outputs. Visual generation produces room floors, character atlases, and objects under a common style specification; assembly connects these assets to the playable geometry and entity identities.

Validation targets four properties: semantic completeness, cross-modal identity, navigable and readable space, and persistent interactive state. Structural validators check references, inventories, asset provenance, and image contracts. Visual inspection and interactive checks supply different evidence about readability and use. When a check identifies an invalid character–item relation, that dependency can be revisited while unaffected stages remain fixed. This instantiates repair as a constrained part of generation, consistent with the broader PCG literature [2].

Running example. Tidal Embassy of Lost Languages turns linguistic scarcity into institutions and actions. Its generated configuration contains eight rooms, 25 named main characters, and 20 item types. Diplomats, smugglers, and memory divers receive different possessions and private knowledge. The setting exposes activities such as translating an archaic bribe, forging a linguistic seal, and rescuing a flooded treaty. These relations create reasons to move, exchange, withhold information, and seek cooperation. Appendix G follows the full generation trace, including failed room images and localized visual repair.

3.3 Bounded Observation and Situated Intention

At round t , an activation policy selects agents. Agent i receives

$$o_i^t = (\ell_i^t, N_i^t, I_i^t, K_i^t, H_i^t, Q_i^t), \quad (3)$$

comprising location, nearby actors, inventory, accessible knowledge, episode and task memory, and selected directed relationships. Role and goals provide the decision context. The observation boundary is intended to preserve information asymmetry, an important distinction from omniscient social simulation [21].

The agent chooses a partner and an intention, using either a generated action or an open proposal. Visible actions explain what an institution makes possible; proposals express intentions absent from that initial vocabulary. Open proposals in the audited implementation support four effect primitives: speech, transfer, item creation, and temporary status. Movement is a separate action route. These primitives can describe new contextual intentions, but they are not arbitrary executable

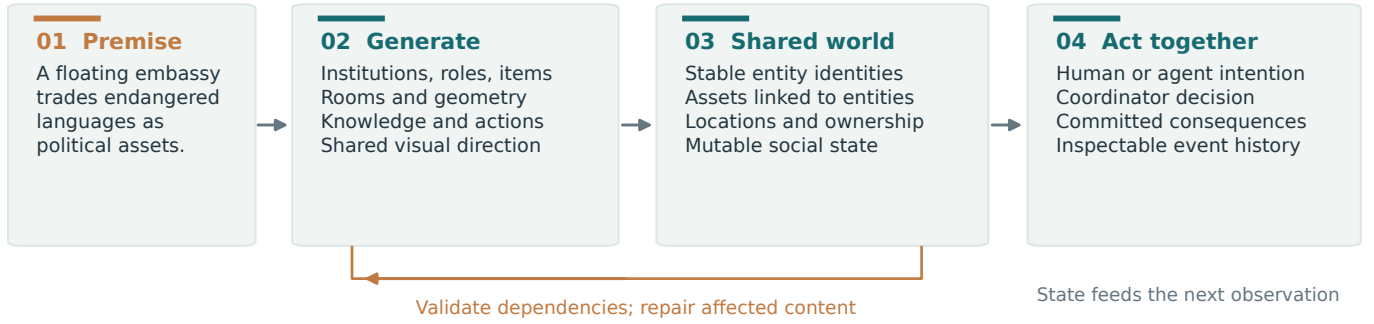


Figure 2. From premise to persistent consequences. Specialized generation stages populate a shared world representation. Validation connects semantic, visual, and spatial requirements; local repair revisits affected dependencies. Humans and agents act through the same world, while logged decisions and state changes support separate construction and interaction evaluations.

Effect	Required grounding
Speech	Addressable participant and a permitted channel
Transfer	Existing item, ownership, quantity, and co-presence where required
Trade	Transfer conditions and the partner’s acceptance
Creation	Declared available inputs and a resulting object identity
Status	Permitted target, bounded magnitude, and duration
Movement	Reachable destination and valid occupancy

Table 2. Examples of action conditions in the shared world. Linguistic plausibility alone does not establish that a requested effect is executable.

programs or a general physics engine. Humans can approach an inhabitant, send messages, use or trade items, select a visible action, or formulate a proposal through the same world interface. This extends the shared grounded-play tradition of LIGHT [12] to premise-generated settings.

3.4 Coordinator-Governed State Changes

We write an open proposal as $q = (i, j, u, F, C)$: actor i , optional target j , language intention u , requested effects F , and declared consent evidence C . The model supplies a structured proposal; the audited coordinator core uses deterministic schema and state checks to return a decision and explanation,

$$\Gamma(W, S_t, q) = (d, r), \quad d \in \{\text{approve, revise, reject}\}. \quad (4)$$

Following the action-grounding role of Concordia’s Game Master [6], interpretation is separated from committing effects. An approved effect is applied by the state-transition machinery. Revision or rejection leaves the requested material effects unapplied, while the attempt can remain in the interaction history.

The resulting loop combines generative interpretation with state constraints (Table 2). Ownership, quantity, location, and duration are represented explicitly. Social interpretation, including the meaning of an exchange or an accusation, remains model-mediated upstream. Declared partner acceptance is an input to validation; it is not itself proof of independently obtained human consent. Natural-language institutions are not automatically formalized as executable invariants by these checks. Consequently, the coordinator is a practical adjudica-

tion mechanism, not a proof of correct interpretation or universally valid social norms. Our evaluation reports observed outcomes rather than claiming exhaustive rule compliance.

4 Evaluation Design

We separate three units of evidence: a generated world, an interaction event, and an action result. A world can pass structural checks without producing engaging play. An event can contain several effects or attempts, so the number of logged events need not equal the number of adjudicated results. Longer trajectories describe the simulated population, while human judgments require a separate instrument and participants.

The studies address four questions: **RQ1**, how reliably can a premise produce a complete world and at what cost; **RQ2**, which generated and open action pathways are actually used; **RQ3**, whether humans are reachable through the common interaction interface; and **RQ4**, what repeated social structure appears in longer historical trajectories. A fifth question asks whether high execution success coincides with material state changes under a longer, scripted policy harness. Table 3 states the scope of each answer.

5 World Construction Results

5.1 Paired Protocol and Completion

The generation comparison uses Gemini 3 Flash Preview at temperature 0.2 and low thinking level for both methods. Five primary premises have three repetitions each; five held-out briefs have one each, giving 20 trials per method. The held-out set and evaluator were frozen before held-out generation. The requests include multi-sentence design constraints; this is not a strict one-sentence-input test (Appendix B). The analysis unit is the shared brief: primary-world binary outcomes use a majority over the three repetitions, while held-out outcomes use the single trial.

The single-pass baseline receives a 24,576-token output cap, three times the 8,192-token cap of one specialized stage. This provides a strong single-call allowance, but does not equalize total computation across methods. The compositional pipeline makes multiple calls and can use local retries. We therefore report first-pass and eventual completion separately, retaining failed outputs. The comparison evaluates the two com-

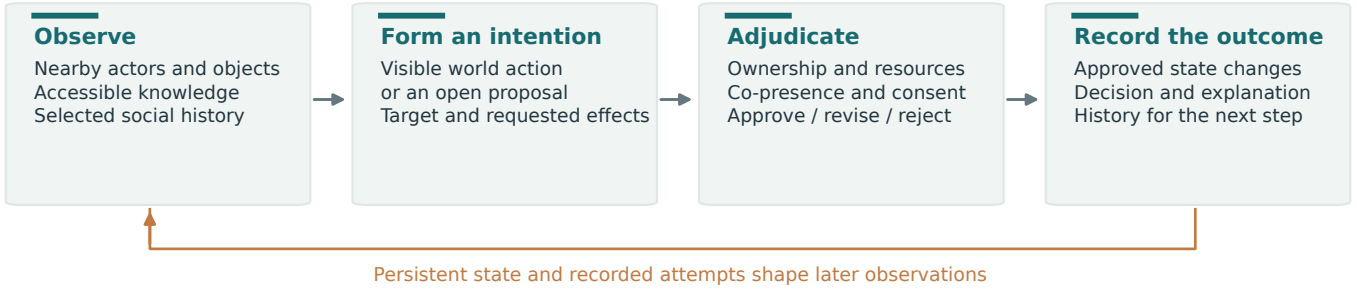


Figure 3. One interaction cycle. Bounded observations inform a visible action or an open proposal. The coordinator evaluates the proposed effects; approved changes update shared state. Both outcomes and failed attempts provide history for later decisions.

Study	Unit and scope	Evidence reported	Interpretation
Generation comparison	10 paired premises; 20 trials per method	Completion, structural quality proxies, cost	Descriptive reliability–cost tradeoff
Validation and repair	58 injected faults with clean checks; 5 repair cases	Detection, attribution, local repair cost	Specified contracts and seeded defects
Multiworld interaction	5 worlds; 2 rounds each; 53 events	Action pathways, dyads, outcomes, state changes	Initial behavior under common runtime settings
Cross-model audit	24 generations; 21 scripted 24-round runs	Completion, executed effects, rejected actions	Retrospective coverage and execution stress test
Historical pilot	3 prior runs; 1,052 events	Repeated ties and relational trajectories	Exploratory, confounded across worlds
Human evaluation	Matched English/Chinese artifact forms	Released materials; no collected ratings here	Prospective artifact judgments
Memory intervention	Planned 3 seed pairs, 32 rounds	Protocol only; no paired results here	Prospective within-world mechanism test

Table 3. Evaluation scope and evidence status. Completed observations are separated from evaluation materials and planned experiments.

World-level result	Compositional	Single-pass
First-pass completion	9/10	5/10
Eventual completion	10/10	7/10
Primary mean tokens/trial	48,527	22,250
Held-out mean tokens/trial	77,556	38,922
Primary median latency (s)	101.2	55.8
Held-out median latency (s)	161.1	86.1

Table 4. Completion is aggregated at the premise level; cost and latency are trial-level summaries within each split. The primary split has 15 trials per method and the held-out split has five.

plete generation procedures; it does not isolate decomposition at equal total token budget. This joint reporting of reliability and cost follows the evaluation concerns in Kapoor et al. [25].

Compositional generation completes 10/10 worlds versus 7/10 for the baseline; first-pass completion is 9/10 versus 5/10 (Table 4). The paired eventual outcomes contain three compositional-only successes and no reverse discordances; the one-sided exact paired test gives $p = 0.125$. For first-pass completion there are five compositional-only and one baseline-only successes ($p = 0.1094$). These small samples support a descriptive improvement, not a statistically established general advantage. Median generation is approximately 1.8–1.9 times faster for the single-pass method, which uses about half as many tokens.

5.2 Quality Proxies, Validation, and Repair

The six frozen quality measures show a mixed profile. Compositional outputs have higher mean merchant-inventory validity, premise-item recall, role uniqueness, room-canon adherence, and wardrobe-canon term coverage. The single-pass baseline has higher premise-agent recall (0.5593 versus 0.4632). The wardrobe proxy favors composition in nine of ten worlds (one-sided sign test $p = 0.0107$, unadjusted across the six measures). These are structural or lexical proxies; a higher score does not establish aesthetic preference or better play. We show all six in Figure 4, including the unfavorable comparison.

The seeded-fault suite mutates copies of five completed artifacts and invokes the production *deterministic validators*, without language-model or image-generation calls. It detects and attributes all 58 injected faults, with no flags in 58 corresponding clean checks. Faults cover missing semantic records, inventories, provenance mismatches, and image/assembly contracts. The clean checks reuse artifacts across categories, so they are not 58 independent worlds. This result concerns the specified fault suite; it is not 58 successful aesthetic judgments or evidence of universal defect detection.

For five injected role–inventory inconsistencies, localized refinement restores all five cases while preserving five unaffected intermediate stages. Mean model-call and token reductions relative to the full-rebuild reference are 48.0% and 39.9%. These observations support repair locality on the seeded failure



Figure 4. Construction evidence at its measured resolution. (A) Paired premise-level completion; the primary split uses majority outcomes. (B) Per-trial token use by split; Table 4 reports latency. (C) Differences in structural and lexical quality proxies, including the metric favoring the baseline. (D) Local-repair resource ratios for five cases. All values come from frozen records; no new model trials were run for this revision.

type. A wider sample of naturally occurring defects is needed to estimate its general benefit.

5.3 Visual and Human-Evaluation Evidence

The five demonstration worlds connect generated maps, characters, objects, and persistent interaction. The Tidal Embassy trace includes rejected room images, local refinement, directional character atlases, and a 25-character interactive trial with movement, messaging, item use, trade, and a model-mediated reply (Appendix G). This is implementation and case evidence, not a blinded comparison of visual quality.

The separate comparative questionnaire supplies blinded, standardized views of actual artifacts, with a model-comparison arm and a compositional-versus- single-pass arm. It evaluates written design, not runtime reliability or participant enjoyment. No collected human ratings are reported here; Appendix D records the prospective scope.

6 Interaction Across Five Generated Worlds

6.1 Protocol and Metric Definitions

The five held-out worlds are Aurora Court of Migrating Cities, Clockwork Rain Conservatory, Mycelium Patent Bazaar, Sunken Satellite Monastery, and Tidal Embassy of Lost Languages. Each contains 24–25 main characters and 11–12 dis-

coverable actions spanning five interaction families. At least five activities per world derive from its premise.

Each accepted run uses Gemini 2.5 Flash for two rounds, activation probability 0.12, and a fixed world-specific seed (26080411–26080415). Agent-profile generation is not repeated during these runs. Image and video generation are disabled, and all five runs pass the world-identity audit. The runtime settings are shared, but the seeds and world contents differ: this is a descriptive multiworld study, not a paired intervention isolating premise effects.

An *event* is a recorded interaction. An observed participant is any actor or target ID, including the human interactor where present. A dyad is an unordered pair of distinct actor and target IDs. If n_{ij} is its event count, the repeat fraction is

$$\rho = \frac{\sum_{\{i,j\}} \max(n_{ij} - 1, 0)}{N_{\text{events}}}. \quad (5)$$

Action entropy is $H = -\sum_a p(a) \log_2 p(a)$ over logged route identifiers. An open event has a `proposed_route`; an explicit world-action event has a `world_action_route`. Other routes can still encode premise-specific activities. Thus these categories identify execution pathways, not an exhaustive semantic classification of world relevance.

Action-result success is computed separately from round

summaries:

$$s = \frac{\sum_{w,t} n_{w,t}^{\text{success}}}{\sum_{w,t} n_{w,t}^{\text{results}}} = \frac{49}{56} = 87.5\%. \quad (6)$$

An event can include multiple effects or attempts. Neither 53 events nor 37 dyads is the denominator of this rate. Appendix A records the per-world numerators and denominators.

6.2 Action Pathways and Consequences

The runs produce 53 events, 37 within-world unique dyads, ten approved open proposals, and nine explicit world-action events (Table 5). Sunken Monastery uses eight explicit world actions, including purging corrupt packets and negotiating bandwidth tithes. Clockwork instead produces seven open-proposal events out of twelve. This contrast shows two available execution pathways in use, without proving that one causes better behavior.

In Clockwork, a jammed weather-sheet press becomes a local investigation. Marius confronts Valerius about the failure; Seraphina asks about etching; Valerius discusses sheet authenticity. In round two, the same participants revisit sabotage, inspect a brass folio, and redirect suspicion. Six actions form this three-character sequence; a seventh open proposal concerns the market impact of the jam. The approved effects update directed relationships while the press failure remains part of shared context. The complete episode diagram is retained in Appendix F.

The resulting state changes differ across worlds. Clockwork accumulates +3 net trust and +21 influence/fear, whereas Aurora and Tidal accumulate +70 and +88 trust with zero net influence/fear. These are simulated state variables produced through the model-mediated interaction machinery. They show that the traces have persistent consequences, but they are not independent psychological measurements or evidence that the premise alone caused the differences.

6.3 Human Reachability and Failed Attempts

Thirteen events target the human interactor across four worlds; Clockwork has none. The logs establish that agents can select the player as a partner within the shared world. They do not establish how often a human replied, whether cooperation succeeded, or whether participants enjoyed the encounters. Those questions require participant-level outcome measures.

Seven of the 56 adjudicated action results are unsuccessful. Retaining these attempts in the history permits later revision, but this study does not measure whether failure feedback improves subsequent success. The evidence supports the availability of an outcome-sensitive interaction loop, consistent with the broader importance of environment feedback in language agents [17–19].

7 Cross-Model Construction and Scripted Execution

7.1 A Separate One-Sentence Corpus

The August 25 archive adds eight recorded model identifiers and three shared one-sentence prompts: a city taxing spoken promises by weight, a railway greenhouse allocating access through crop-care obligations, and a courthouse rebuilding a shoreline from confiscated objects. This later pipeline requests

12 agents and four player slots. We re-audit its 24 generation outcomes separately from the earlier ten-brief comparison: the prompt set, implementation, and available models changed, so their scores are not pooled.

Seventeen of 24 trials yield complete artifacts, with nine first-pass completions. Five models complete all three prompts; Gemini 3.5 Flash Lite and GPT-5.6 Terra each complete only the first, and GPT-5.6 Luna completes none. Table 7 retains every failure. These are recorded API identifiers, not guarantees of immutable provider snapshots. One generation per model–prompt cell supports a coverage diagnosis, not a stable model ranking. Although the archive calls the suite hidden/confirmatory, it lacks a pre-execution immutable registration, and some scoring and retry policies were revised after development runs. We treat it as retrospective evidence.

7.2 What the Longer Harness Actually Measures

Seven first-prompt artifacts support three repeated 24-round trajectories each. The harness makes six calls per trajectory, at rounds 1, 5, 9, 13, 17, and 21; each call plans 12 actions over four rounds. Its planner observes summaries of *all* agents and recent events. It is a centralized multi-character policy, distinct from the locally observed agents in Section 3. Scripted requests, an invalid transfer, a resource shock, and a cross-room request probe execution. The prompt explicitly asks for thread continuation, legal recovery, and human follow-through; these behaviors cannot be claimed as spontaneous emergence or participant engagement.

The 21 trajectories contain 1,512 attempted model actions, of which 1,365 succeed (90.3%). Among successful actions, 398 (29.2%) record an inventory or location mutation. Among 314 approved proposals, 150 (47.8%) declare at least one non-speech effect; status-only effects count in this latter measure. These denominators address different questions and are not interchangeable. Figure 6 reports model-specific counts and repeat ranges instead of a weighted quality score.

7.3 Failure Modes and Evidence Boundaries

Gemini 3.7 Flash succeeds on all 216 attempts, but only 19 successes change inventory or location and only four of 22 approved proposals have non-speech effects. Terra’s 18 approved proposals are all speech-only, despite 214/216 successful actions. Conversely, Gemini 2.5 Flash has 42 failed attempts but 34/45 approved proposals with non-speech effects. These observations motivate reporting *what changed* alongside whether an action executed, consistent with analytical trajectory evaluation [24].

All 147 failed model actions and all 21 scripted invalid transfers leave their recorded state hashes unchanged. Final-state serialization round trips match in 21/21 runs; this checks serialization, not recovery after a process crash. Of 1,021 recorded relationship updates, 1,013 have the same $(\Delta\text{trust}, \Delta\text{affection}, \Delta\text{fear}) = (4, 2, 0)$. The custom-action handler supplies exactly this default for non-aggressive actions. Thus rising relationship values in this harness largely reflect execution rules rather than independently demonstrated social development. Appendix B provides the full audit and replication boundaries.

World	Events	IDs	Dyads	Repeat	H (bits)	Open	World route	Human-targeted
Aurora Court	9	12	7	22.2%	1.447	1	1	2
Clockwork Conservatory	12	8	6	50.0%	3.252	7	0	0
Mycelium Bazaar	9	12	7	22.2%	1.392	0	0	4
Sunken Monastery	11	13	9	18.2%	1.823	1	8	3
Tidal Embassy	12	8	8	33.3%	1.208	1	0	4
Total	53	–	37	–	–	10	9	13

Table 5. Two-round interaction records. IDs count observed participants, including the human where applicable. Open and world-route counts follow logged route prefixes; other routes may also be premise-specific. Dyads are unique within each world.

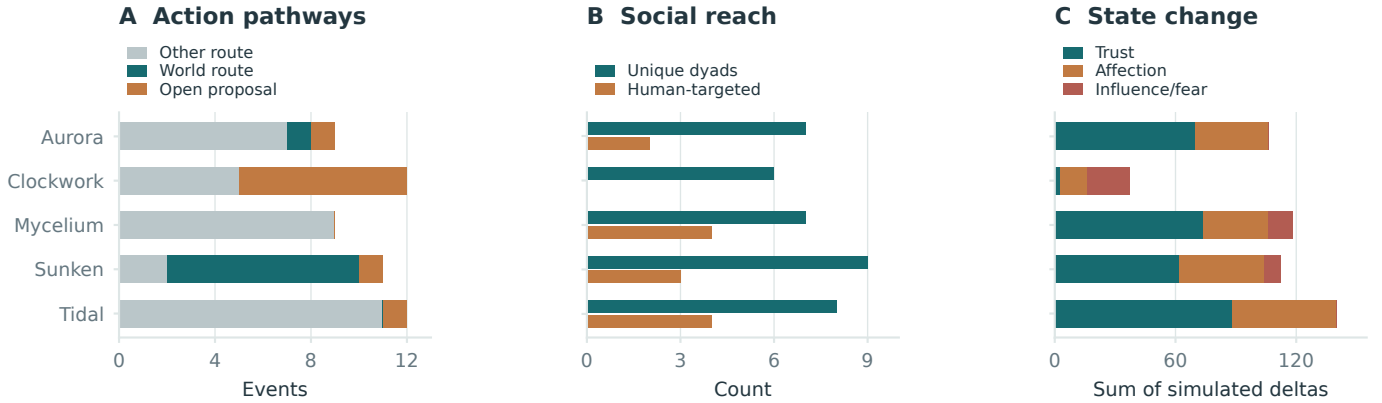


Figure 5. Interaction profiles under common runtime settings. (A) Logged action pathways. (B) Unique dyads and events targeting the human interactor. (C) Sums of model-mediated relationship changes. The five worlds have one short run each; bars are descriptive totals, not population estimates.

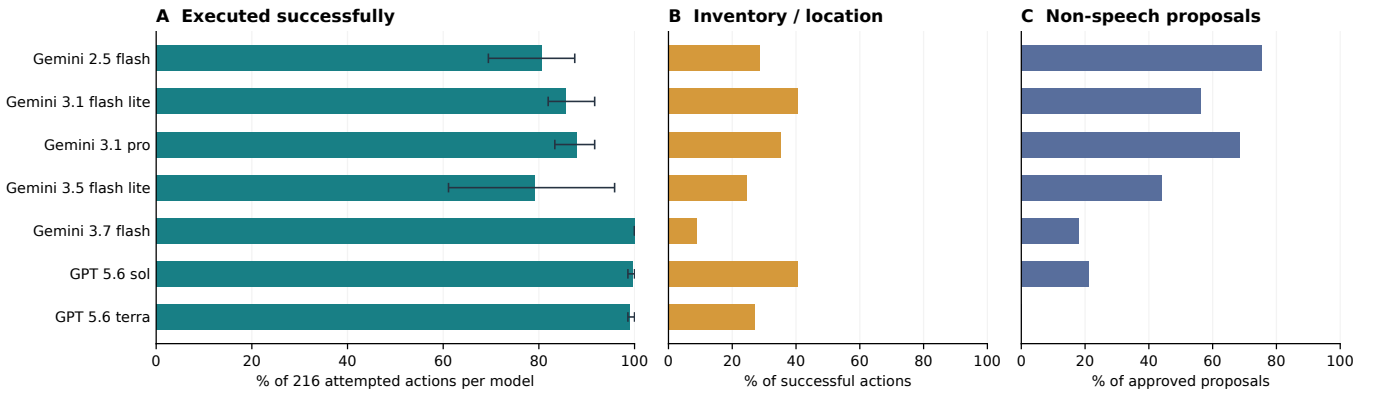


Figure 6. Successful execution does not ensure rich consequences. Seven generated worlds, each with three 24-round centralized-policy runs. (A) Success per attempted action; whiskers are the minimum and maximum over three runs, not confidence intervals. (B) Inventory/location mutations among successful actions. (C) Non-speech effects among approved proposals, including status changes. All comparisons couple a model with its own generated world; they do not isolate policy capability on identical environments.

8 Historical Context

Three earlier runs add 1,052 events across an adventuring guild, a black-market auction, and a storm-struck cruise. Between 23.4% and 36.5% of events revisit an existing dyad, while the largest connected components contain only 11–13 participants. These records motivate studying communication across local clusters, but configurations differ and spatial scheduling can produce the observed structure. Full tables and trajectories appear in Appendix C; they are not pooled with the later scripted harness.

9 Discussion and Limitations

A contribution at the connection between authoring and simulation. Agora connects staged world authoring, explicit persistent state, and grounded social action. Its closest predecessors already establish language-driven game construction [3, 4], shared human-agent play [12], and Game Master adjudication [6]. The contribution is their integration around a premise-conditioned world representation and inspectable trajectories. The comparison here is against a single-pass generation baseline, not those full systems; no cross-system superiority is established.

Discoverability and expressiveness are separate design needs. A generated repertoire explains what inhabitants can do in a new setting. Open proposals let a particular encounter depart from that vocabulary. The *Sunken* and *Clockwork* traces illustrate both pathways. Their presence does not establish the benefit of openness: a matched proposal ablation, with equivalent world and agent states, is needed to estimate that effect. Likewise, interface access does not establish useful or enjoyable human collaboration.

Construction evidence has limited scope. Ten shared briefs provide a small sample, and composition receives more total computation. Structural checks and lexical proxies do not establish aesthetic preference. The reported statistical evidence remains exploratory. Human ratings, uncurated premises, budget-matched comparisons, and naturally occurring failure cases would strengthen the construction evaluation.

Execution is not behavioral validation. The five locally observed runs last only two rounds with one seed each. The 21 longer runs use centralized block planning, scripted human events, and each model’s own generated world. Their high execution success does not validate emergent coordination; identical relationship increments expose a contribution from default state-update rules. Historical worlds are confounded by prior configurations. Relation deltas are model-mediated state variables, and route entropy depends on the action vocabulary. Neither measures human trust, fear, affection, or social intelligence directly. LLM judging also carries systematic biases [23]; behavioral fidelity would require independent human data and explicit validation targets, as in human-grounded agent studies [27].

A prospective intervention is not a completed result. The Tidal protocol varies access to prior social episodes while preserving current world state. No completed paired 32-round results or blinded annotations are reported in this release. The available record contains a local protocol, but no external registration identifier; we therefore describe it as a prospective design. Three seed pairs would provide a feasibility estimate with substantial uncertainty, not precise population inference.

Reproducibility and human participation. Hosted models may change despite a recorded model name; available settings, seeds, exclusions, and frozen evidence are documented in Appendix A. Human studies should evaluate refusal, outcome legibility, and collaboration under unequal information. Fictional roles do not validate demographic models.

10 Conclusion

Agora turns a short premise into a rendered social environment whose inhabitants can act on persistent shared state. The completed studies show an observed reliability–cost tradeoff in generation, use of explicit world actions and open proposals, and a gap between successful execution and material consequences in a separate 21-trajectory audit. The central result is an inspectable connection between world authoring and consequential interaction. Broader human evaluation and controlled interventions are needed to determine how well the generated worlds support collaboration and which mechanisms shape their longer social trajectories.

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World	Successes	Results	Rate
Aurora	9	9	100.0%
Clockwork	11	13	84.6%
Mycelium	8	10	80.0%
Sunken	11	11	100.0%
Tidal	10	13	76.9%
Total	49	56	87.5%

Table 6. Audited action-result counts, summed directly from the accepted story bundles’ round summaries. The 61-result denominator in the previous manuscript was a transcription error.

A Reproducibility and Evidence Audit

Generation settings and aggregation. The frozen protocol uses `gemini-3-flash-preview`, temperature 0.2, and low thinking level. Specialized stages have an 8,192-token output cap; the monolithic call has a 24,576-token cap. Five primary premises have three repetitions each and five held-out premises have one each. A primary premise passes a binary completion endpoint when a majority of its repetitions pass; the five held-out premises use their single observed outcome. Costs summarize all trials within each split. Failed outputs are retained. Quality-evaluator definitions and the held-out set were frozen before held-out generation.

The manuscript’s construction figures read the frozen file `paper_evidence_summary_20260729.json`. The six quality proxies are merchant-inventory pass rate, premise-item recall, premise-agent recall, role uniqueness, room visual-canon adherence, and wardrobe-canon term coverage. They are reported as proxies and are not substituted for human judgments. The seeded validator report is included alongside these summaries.

Accepted interaction runs. The accepted records are `mw3_aurora`, `mw3_clockwork`, `mw3_mycelium`, `mw3_sunken`, and `mw3_tidal`, collected on August 4, 2026. They use Gemini 2.5 Flash through AI Studio, two rounds, activation probability 0.12, and seeds 26080411–26080415 in that order. Compiled profiles are reused, and image/video generation is disabled. The archived run configuration does not by itself establish an immutable provider model snapshot; decoding details absent from that record are not inferred.

Earlier `mw_` attempts are excluded because a shared mutable scenario directory caused cross-world contamination. Earlier `mw2_` attempts made no model progress because DNS was unavailable. The accepted runs place mutable scenario state in separate directories and pass the world-identity audit. These exclusions concern execution validity and are distinct from excluding an unfavorable behavioral outcome.

Success-count correction. Table 6 distinguishes action results from story events. The frozen metrics already reported 0.875; summing the raw round summaries recovers 49 successes from 56 results. This revision corrects the prose denominator without changing the recorded success rate or event count. The accompanying audit file includes the source paths, SHA-256 hashes, and extracted round counts so that this correction is traceable.

Historical analysis. The pilot records are the guild run of April 22, 2026, the black-market run of April 19, and the cruise run of April 18. They are analyzed independently of the August interaction runs and the planned intervention. Observed-ID counts are 103, 99, and 100, respectively; these are counts of recorded actor/target identifiers, not a claim that all runs had an identical realized population.

Artifact scope. The release contains the paper sources, BibTeX database, figure programs, frozen figure inputs, evidence-audit summaries, and a separate revised bilingual comparative-questionnaire package. The bundled inputs reproduce the figures and reported summary arithmetic; full runtime replication additionally requires the original implementation, configuration, model access, and raw execution records. The expanded release reanalyzes existing August 25 generation and execution records. No new model call, human response, or paired memory-intervention result was produced during this revision.

B Extended Evidence: Counts and Harness Audit

Source selection. Generation outcomes come from the 24 result records under the archive’s `confirmatory_runs` directory. A first-pass completion is a complete artifact with no recorded localized generation retry. Retry lists for some failed cases were not preserved in the original format; we therefore do not estimate total repair incidence from those lists. Failed cases remain failed. The 21 execution sources are the corrected `runtime_trajectories_v2/replicate_*` records. Earlier runs affected by human placement and transport defects are not substituted or pooled. All raw result and trajectory records used here are frozen with SHA-256 hashes and recomputed by the bundled audit script.

Input scope of the earlier construction comparison. The ten requests in Section 5 are *world briefs*: some include several sentences specifying institutions, cast size, room requirements, and visual style. They do not test strictly one-sentence input. For the comparative questionnaire, the archived single-pass prompt contains an original-request JSON exactly matching the corresponding compositional request for all ten selected first replicates. This confirms a shared user request; the method-specific system instructions and total computation differ. The three later prompts are separate, genuinely one-sentence semantic inputs.

Execution schedule and observability. Each four-round block is planned from one snapshot; intermediate actions can make later planned actions invalid. At each checkpoint the policy receives room connectivity, all-agent summaries, the action catalog, up to four open action examples, the current intervention, and the 24 most recent events. Decoding uses the recorded temperature 0.75 and low thinking setting. There is no archived claim of matched random seeds across providers. A model generates and controls its own world, so world difficulty, generation quality, and policy quality are confounded. The harness is an executable-plan stress test, not an information-asymmetric social experiment.

Recorded model	Complete	First pass	Executed	Inventory/location	Approved	Non-speech
Gemini 2.5 Flash	3/3	2/3	174/216	50/174	45	34/45
Gemini 3.1 Flash Lite	3/3	2/3	185/216	75/185	89	50/89
Gemini 3.1 Pro Preview	3/3	0/3	190/216	67/190	19	13/19
Gemini 3.5 Flash Lite	1/3	1/3	171/216	42/171	102	45/102
Gemini 3.7 Flash	3/3	3/3	216/216	19/216	22	4/22
GPT-5.6 Luna	0/3	0/3	N/E	N/E	N/E	N/E
GPT-5.6 Sol	3/3	1/3	215/216	87/215	19	4/19
GPT-5.6 Terra	1/3	0/3	214/216	58/214	18	0/18
Total	17/24	9/24	1365/1512	398/1365	314	150/314

Table 7. Retrospective counts without reviewer scores or rank ordering. Construction columns use three prompts per model. Execution columns use only the first prompt’s generated artifact and three repeated trajectories. N/E means no eligible artifact; it is not a runtime zero or a human preference. “Non-speech” includes status effects and is counted from approved proposals.

Scripted human and recovery probes. The round-5 request and round-21 follow-up are supplied by the harness. When needed, the harness moves the human into an occupied room before a request or rejection probe; this movement is not counted among model actions. The round-9 probe requests transfer of an item the human does not possess. All 21 such probes are rejected without a state-hash change. A resource shock at round 13 and a cross-room instruction at round 17 are likewise scripted. Target acceptance in model-generated proposals is a declared field, not an independently elicited human response.

What counts as a consequence. The inventory/location endpoint includes successful movement as well as inventory effects, and therefore is not a measure of economic transformation alone. The non-speech endpoint includes temporary statuses and describes the requested effects of approved proposals; approval does not establish that every intended semantic consequence occurred. The 147 failed model actions contain both before/after hashes and a false state-change flag. Five references to nonexistent event IDs occur in failed Gemini 2.5 Flash actions; no successful action in the archive has such an unknown reference. Thread labels and event references are supplied by the planner, so retaining them is not independent proof of causal coherence.

Statistics and reproducibility. The independent scope is 24 generation cells and 21 trajectory records, not 1,512 independent experimental units. We report counts and the observed range over three runs, without asymptotic confidence intervals, significance stars, or a composite leaderboard. The audit reads raw outcomes and mutations and does not load the identity-visible expert review or its weighted scores. The source snapshot also includes the planner harness, coordinator validation, and custom-action handler used to interpret the records. Source inspection supports the mechanism explanation but does not replace fresh execution tests against an immutable historical checkout.

C Exploratory Historical Trajectories

We separately analyze three prior runs configured around approximately 100 agents: an adventuring guild (25 rounds), an underground black-market auction (20), and a storm-struck

luxury cruise (20). They predate the current hypotheses and differ in configuration and social context. We reconstruct undirected actor–target networks, route entropy, and sums of directed relationship changes using the same metric definitions.

Between 23.4% and 36.5% of events revisit an existing dyad, while each world contains 229–282 unique dyads (Table 8). Route entropies are 2.889, 2.619, and 2.524 bits for guild, black market, and cruise, respectively. Because the repertoires differ, these values describe each trace’s variety; they are not directly comparable measures of social intelligence.

The largest components contain only 11–13 participants despite many observed IDs. Figure 7 shows dyads continuing to accumulate after component sizes stabilize. This is consistent with additional ties forming predominantly within local clusters. Spatial encounter and activation policies can generate such structure, so its social interpretation must remain conditional on those mechanisms.

In the black market, influence/fear (+1194) exceeds trust (+179) and affection (+9); the guild and cruise instead accumulate mostly trust and affection. The pattern is compatible with each scenario’s institutions, but prompts, action definitions, schedules, and run histories are confounded. It motivates a controlled question rather than answering one. In particular, warning diffusion and collective rescue need separate outcomes: receiving information need not produce action [28]. Appendix E specifies a paired memory intervention to examine that distinction.

D Prospective Comparative Human Evaluation

The revised, separately packaged questionnaire compares written world designs under blinded source labels. One arm pairs successful outputs of different recorded models under the same one-sentence prompt (17 artifacts, 41 pairs). A second arm pairs compositional Agora with single-pass outputs under ten identical archived requests, using the first replicate of each method. Source fields, ordering, item quotas, and truncation rules are fixed across conditions; no summary is rewritten to favor one output. Unavailable model artifacts are recorded as generation failures, not imputed preference losses.

Each participant reviews two pairs per arm. Presentation order and A/B orientation are balanced in an 82-code schedule; this is a coverage plan, not a power calculation. The primary

World	Rounds	Events	IDs	Dyads	Repeat	Largest comp.	Dominant state changes
Guild	25	444	103	282	36.5%	11	Trust +3473; affection +1745
Black market	20	299	99	229	23.4%	13	Influence/fear +1194
Storm cruise	20	309	100	235	23.9%	13	Trust +2576; affection +961

Table 8. Exploratory statistics from 1,052 historical events. IDs are distinct logged actor/target identifiers, not a fixed population denominator; the guild record contains 103. Relationship totals are simulated state changes.

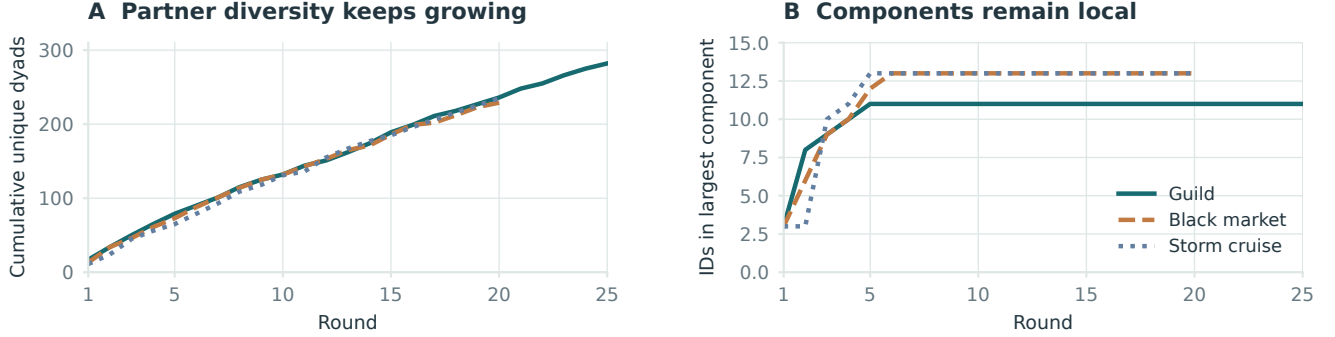


Figure 7. Longer traces distinguish activity from connectivity. Unique dyads continue to accumulate, while the largest observed components plateau early. These historical runs motivate testing transmission across local clusters; they do not identify its causal mechanism.

question asks overall design preference. Five separate diagnostic items address premise realization, rules and activities, social connections, places and objects, and possible actions. Each permits a directional preference, a tie, or insufficient evidence. Comments and quotation permission are optional. Instructions are bilingual, while source evidence remains in its original English, restricting recruitment to readers of English. No cross-language equivalence claim is made.

The prospective analysis reports prompt-specific outcomes, ties, abstentions, and participant-clustered intervals conditional on the fixed corpus. It does not treat repeated ratings as independent participants or aggregate different constructs into a validated scale. The unequal-compute architecture comparison estimates preferences for the observed system configurations. No matching baseline images exist, so image quality and actual play experience are outside the comparative instrument. Clear construct and reporting boundaries follow human-evaluation guidance [29]. No participant count, preference, agreement estimate, or completed human-study result is claimed in this paper.

E Prospective Tidal Memory Intervention

Status. This appendix describes planned work, not an additional completed experiment. The project contains a local protocol and a one-round full-memory smoke run; it does not contain completed paired 32-round results supporting a causal memory claim. No external preregistration identifier is available in the release. The changes to analysis and measurement specified below must be frozen before confirmatory runs and are not represented as an already registered plan.

E.1 Intervention and Hypothesis

The 25-agent Tidal Embassy world is initialized once and reused. At round zero, three agents in the lower archive privately learn that a flood threatens the only surviving peace

treaty unless it is moved and authenticated. The full-memory condition retains prior social episodes. The ablation removes prior-round social episodes from decision context while preserving current locations, inventories, public facts, task state, and directed relationships. Thus it tests access to episodic social history, not removal of all memory: tasks, current knowledge, and relationships can also carry historical signals.

The planned seeds are 26080311, 26080312, and 26080313, with 32 rounds per condition. Initial profiles, world configuration, model and decoding settings, and activation schedule are matched within each pair. Activation draws should be precomputed with a separate random stream; reusing a seed alone does not guarantee matching if the conditions consume randomness differently. API nondeterminism remains a source of variation.

The hypothesis is that access to earlier social episodes increases warning diffusion and coordinated rescue within this generated setting. Generative Agents motivates studying memory in community behavior [5]; the separation of information reach and collective action is also informed by contagion research [28].

E.2 Outcomes and Measurement Controls

The primary endpoint is warning reach by round 16 among the 22 initially uninformed agents. Let $Z_i(16)$ indicate that the agent has expressed or recorded a prespecified treaty-threat proposition by that round. Then

$$D_{16} = \frac{1}{22} \sum_{i \notin I_0} Z_i(16), \quad |I_0| = 3. \quad (7)$$

Measurement must use an append-only audit stream outside the ablated decision memory; otherwise deleting memory could mechanically reduce measured reach even after successful transmission. The audit record is not fed back to agents. Initial warning availability must be preserved equally through its first intended use, so the ablation does not inadvertently remove the treatment trigger itself.

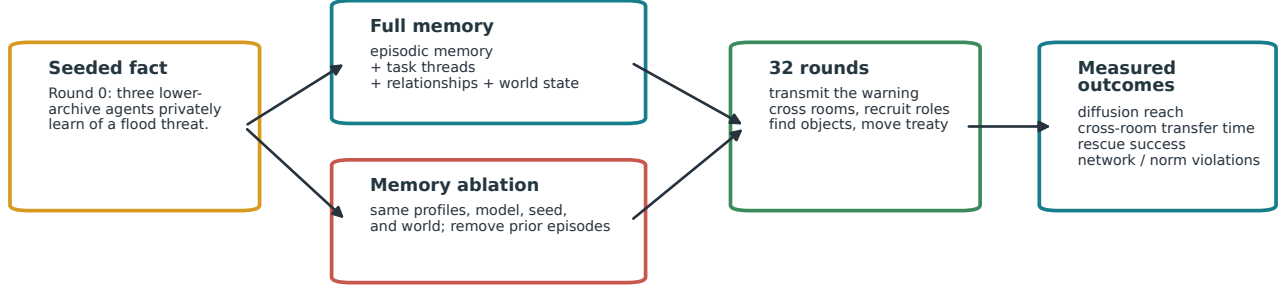


Figure 8. Prospective design; no paired results reported. Matched initial states receive the same private warning. Conditions differ in access to prior social episodes. Diffusion and material rescue are distinct outcomes. The planned unit of comparison is a seed pair.

Secondary outcomes are first cross-room transmission time, rescue-task participation and role diversity, treaty relocation and authentication, unique and repeated dyads, largest component, object transfers, relationship changes, and rule violations. Unreached transmission or rescue events are reported as censored at the horizon. Two condition-blind annotators should label candidate episodes under a frozen proposition codebook; agreement and resolved disagreements must be reported before condition-level interpretation.

Manipulation checks verify that only the designated agents initially receive the warning, that paired initial states and activation schedules match, and that the ablation removes the intended history without exposing hidden facts through summaries or coordinator feedback. These checks implement the information-boundary concern highlighted by Zhou et al. [21].

E.3 Analysis and Precision

The independent comparison unit is a seed pair, not an event or an agent. For seed s , report $\Delta_s = D_{16,s}^{\text{full}} - D_{16,s}^{\text{ablation}}$, both condition trajectories, and the material rescue outcome. Report all three differences and their mean and range. Three pairs are too few for a stable bootstrap confidence interval or a well-powered confirmatory test; additional seeds and a prospectively justified sample size are needed for precise inference.

Qualitative selection rules are the earliest cross-room transmission, first multi-role rescue, largest trust reversal, and first failed coordination. Selections should be applied identically to both conditions and report when an event does not occur. The existing one-round smoke run is execution validation only and must remain outside these comparisons.

F Clockwork Episode

The diagram follows the short episode described in Section 6. It is an illustrative trace from one two-round run, not an independently replicated behavioral effect. Six actions form the central three-character sequence; the seventh concerns the market impact of the malfunction.

G Tidal Embassy: A Generation and Play Trace

We now follow *Tidal Embassy of Lost Languages* from premise to play. Its one-sentence premise describes a floating embassy where linguists, smugglers, ambassadors, and memory divers trade endangered languages as political assets. The example preserves intermediate images and refinement attempts, allowing each generative stage to be connected to the final playable world.

Step 1: Premise analysis and planning. The planning stage identifies the nouns that become state—embassy, languages, translation rights, archives, smugglers, diplomats, and tides—and the verbs that imply gameplay—trade, translate, forge, authenticate, rescue, and reveal. It then proposes an institutional loop: access to archives produces linguistic objects; translators and forgers can translate or forge them; diplomats exchange rights and debts; customs officers can inspect transfers; flood events threaten stored knowledge. The resulting world plan connects entities, conflicts, resource cycles, and open questions.

Planning establishes what must exist and why it matters. Later stages determine spatial layout, visual appearance, and resource quantities. Missing activity loops are refined before visual generation; for example, translation rights must have a source, a use, and a means of transfer.

Step 2: Semantic structure. Semantic generation expands the plan into rooms, roles, items, institutions, knowledge, and conflict hooks with stable identities. In this world, the graph contains eight rooms, including the Phoneme Exchange, Mnemonic Garden, Embassy Customs and Quarantine, and Ballast Hold; 25 named main characters; and 20 distinct items. Explicit relations connect each character to a home, each possession to an owner, and each action to relevant role, knowledge, location, or resource conditions.

The generation process also preserves asymmetry. Shen Mo is a Phonetic Smuggler and Lexical Restorationist based in the Cartographer’s Nook. His inventory includes a translation ear-cuff, archive key, forged exchange ledger, dive helmet, ritual ink, and a fragment of the Unspoken Treaty. His private knowledge includes phonetic decay patterns and ventilation shafts



Subset totals above; the full 12-event run has +3 trust and +21 influence/fear.

Figure 9. An audited two-round episode in Clockwork Conservatory. Six open actions form the central Marius–Valerius–Seraphina sequence; Lyra adds a seventh proposal about market effects. The seven-proposal subset has trust –29, affection –5, and influence/fear +17; the full run has different totals because it includes five other events.

linking Customs to the Mnemonic Garden. Another character receives different property, authority, and information. This uneven allocation is what later makes persuasion, trade, secrecy, and coalition formation possible.

Step 3: Visual direction and cross-modal identity. The visual direction converts semantic identity into a shared canon: Art Nouveau diplomatic architecture, floating platforms, brass infrastructure, bioluminescent language media, water intrusion, and a melancholic ritual tone. It also emits per-entity appearance anchors. Room prompts inherit the canon but add function-specific materials; character prompts inherit world palette and wardrobe rules but retain role silhouettes; props receive icons whose shapes communicate their use. Shared style and entity-specific appearance jointly keep rooms, inhabitants, and objects recognizable as parts of the same world.

The visual direction guides each image independently while maintaining common architecture, materials, palette, and role silhouettes. A visual inconsistency can therefore be refined locally while the semantic identity of the world remains stable.

Step 4: Evolving world state. This stage determines what can change after the images are produced. It creates wallets, inventories, property libraries, knowledge assets, directed relationships, task threads, room occupancy, and world-event status. It also defines invariants. Item quantities remain nonnegative; physical transfers conserve quantity; private knowledge has an owner and an acquisition event; a character occupies one reachable coordinate; relationship changes are directed. The generated repertoire and open-proposal forms are linked to the same world state.

The evolving state connects generated descriptions through observable causes and consequences. The treaty fragment in Shen Mo’s inventory can later be inspected, given away, stolen through an allowed rule, or used as an input to another action. When ownership changes, the next observation shows the new owner.

Step 5: Integrated world representation. The integrated representation connects semantic, visual, spatial, and evolving state. It supports cross-domain questions: Does every home room exist? Does every main character have a visual asset and a walkable placement? Is an item used by an action defined and obtainable? Does a private fact mention an existing institution? Does a room’s purpose agree with its objects and

activities? These dependencies guide local refinement.

If a character image is inconsistent, roles, rooms, and items remain fixed. If an item relation is missing, only the connected content is regenerated. Figure 11 illustrates this process: two room images are refined while the rest of Tidal Embassy remains unchanged.

Step 6: Visual generation. FLUX generates the map source, individual room floors, characters, and local objects from the shared visual direction. Character images are arranged into directional movement atlases, and room images are placed within the planned regions and connected by traversable corridors. Figure 10 shows the composed map alongside three source floors, three character atlases, and an object.

Functional visibility connects visual quality to use: room imagery must cover its playable area, characters must remain recognizable while moving, and furniture must preserve navigable paths.

Step 7: Visual and spatial evaluation. The evaluation examines semantic completeness, visual identity, traversable geometry, foreground coverage, directional character consistency, readable room composition, and visibility during play. The Diplomat’s Lounge required three real generation attempts. Attempt 1 was classified as a poster card with an edge blank-background ratio of 0.56. Attempt 2 removed the card appearance but retained a blank-margin failure with a maximum blank-row ratio of 0.73. Only the third, edge-to-edge playable floor was accepted.

In Agora, visual quality combines coherent identity, readable composition, and fitness for navigation and interaction. Preserved refinement attempts make the progression from attractive illustration to usable world space observable.

Step 8: Persistent play. The generated world begins from a fixed initial state and evolves independently for each session. Humans and AI agents encounter the same rooms, inhabitants, objects, actions, and consequences. Separate world identities keep concurrent multiworld experiments isolated.

Figure 12 connects character generation to embodied play. A crop of the interaction capture shows the selected agent. A complementary world view shows neighboring agents among generated furniture and room surfaces, while the directional atlas records appearance across locomotion. The interactive trial covers movement, messaging, item use, trade, continued play,

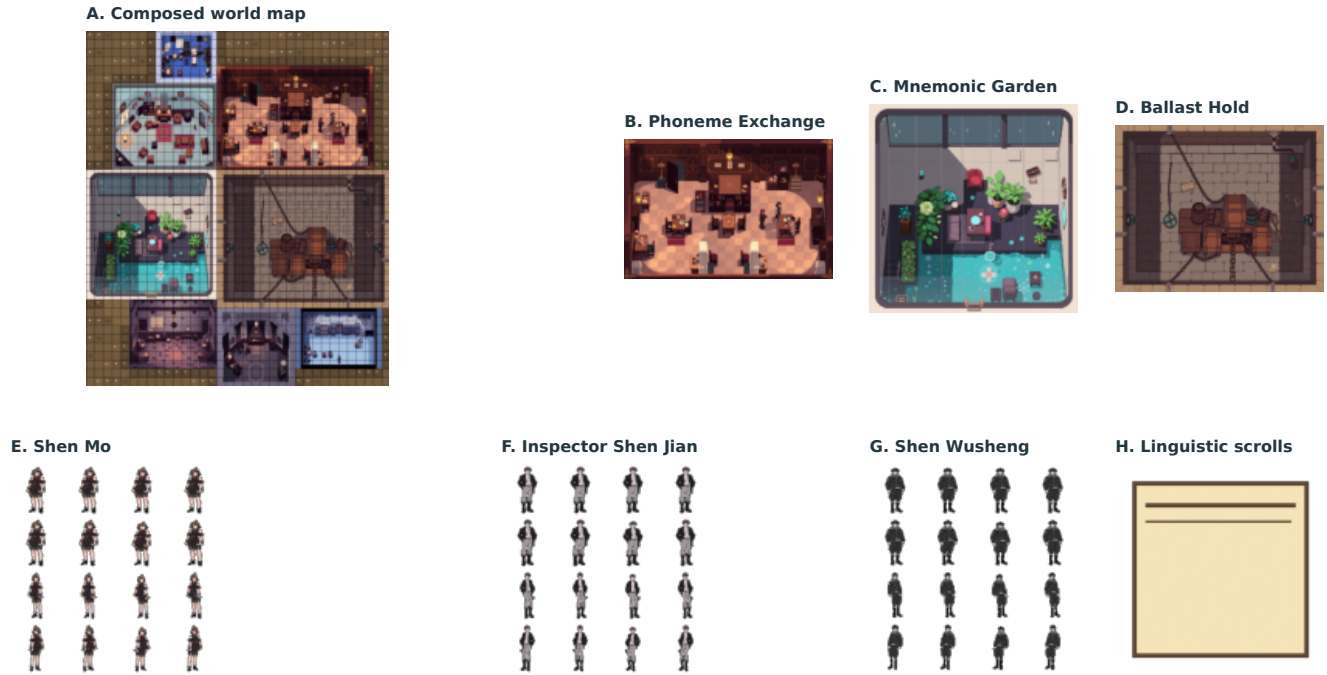


Figure 10. Generated visual elements from Tidal Embassy. The composed map contains eight generated floors (A–D show the map and three constituent rooms); E–G show the directional character atlases; H is a generated object. Shared visual motifs connect spatial, character, and object identity.



Figure 11. Localized visual refinement for the Diplomat’s Lounge. The first image has a large poster-like margin; the second retains an extensive blank boundary; the third fills the playable floor while preserving the room’s identity. Other rooms, characters, and world state remain unchanged.

and a model-mediated reply.

The 25-character set uses one canonical visual identity per inhabitant. Front locomotion and side views preserve the same hair, clothing, body type, and palette, so an inhabitant remains recognizable while moving. Native 64×64 frames retain facial, clothing, and limb detail in the stored atlases.

Character evaluation covers single-person composition, silhouette readability, palette contrast, directional identity, absence of text and detached effects, and visibility against the room surface. Contact-sheet review additionally examines the complete cast together, revealing cross-view identity changes and visual outliers that are difficult to judge one frame at a time.

Successive review passes identified characters with problematic props, enclosing effects, orientation, text, detached

sparkles, or insufficient contrast. These counts reflect different review stages rather than disjoint character sets. Localized regeneration produced a final set of 25 coherent, readable character atlases, all visible during the interactive trial.

Step 9: Interaction-informed refinement. During play, agents and humans observe bounded state and choose either a world-specific action or an open proposal. The World Coordinator evaluates effects and enacts successful transitions, as detailed in Section 3. Interaction traces also provide feedback upstream. An action that is never selected may be poorly named or irrelevant; an unreachable character suggests placement repair; an important object that never enters an exchange suggests a missing action; frequent open proposals around one intention suggest a useful addition to the visible repertoire. Ob-

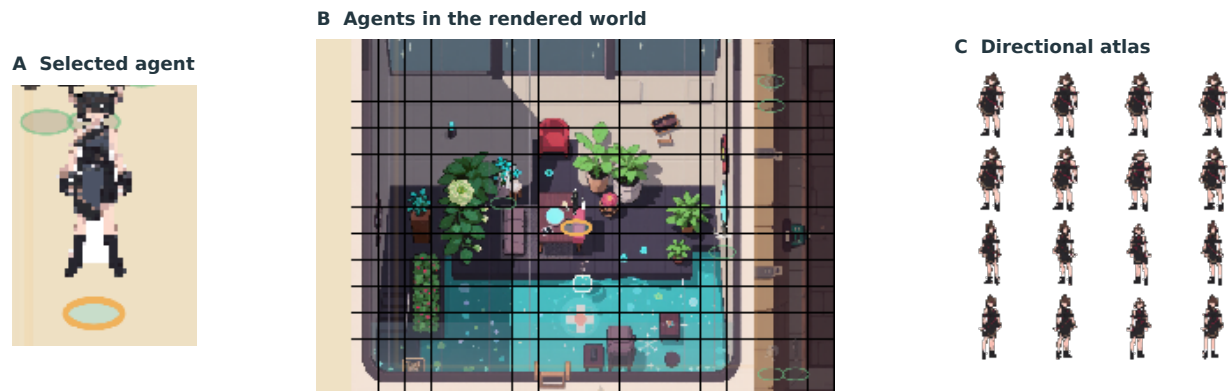


Figure 12. Character identity in Tidal Embassy. (A) A crop of the interaction capture shows the selected character, marked by an amber ring. (B) A complementary world view situates agents among generated room surfaces and furniture. (C) The directional atlas preserves clothing, palette, and silhouette across movement. The interactive trial includes movement, messaging, item use, and trade.

served life inside the world can therefore guide the next round of world refinement.